Theories of hypothesis testing
and
inference for regression

Overview

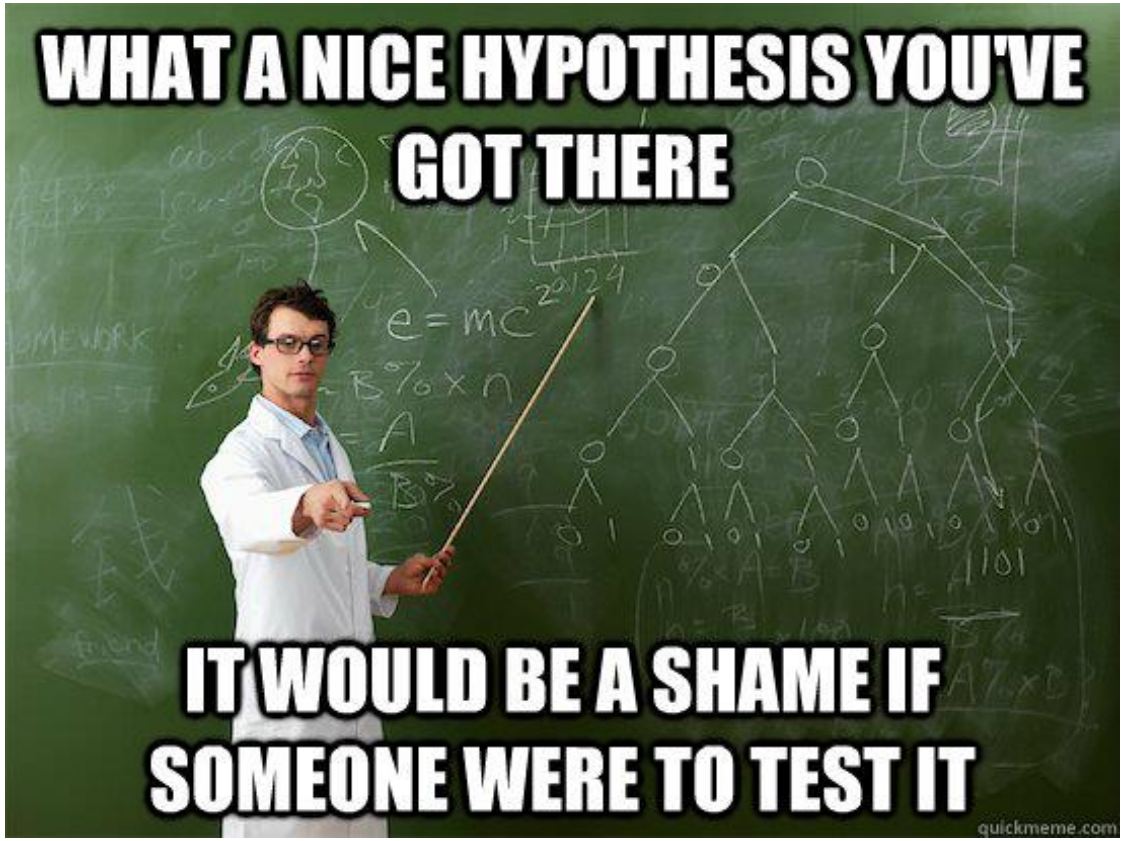
Review and continuation of one-way analysis of variance (ANOVA)

- What does the F-statistic measure?

Theories of hypothesis testing

If there is time

- Parametric inference for regression



Announcement

Final exam review session

- Monday April 27th from 1 pm
- In this classroom (Marsh)

Final exam:

- May 5th (Monday) at 7pm
- Location: Marsh (this classroom)

Final project due Sunday April 26th

If you have accommodations, please reach out to SAS as soon as possible to schedule to take the exam with them

Announcement

No office hours on Friday

- I can hold additional office hours after the review session next Monday from 2:30-3:30 pm.

Download any data you want to save from the YCRC server

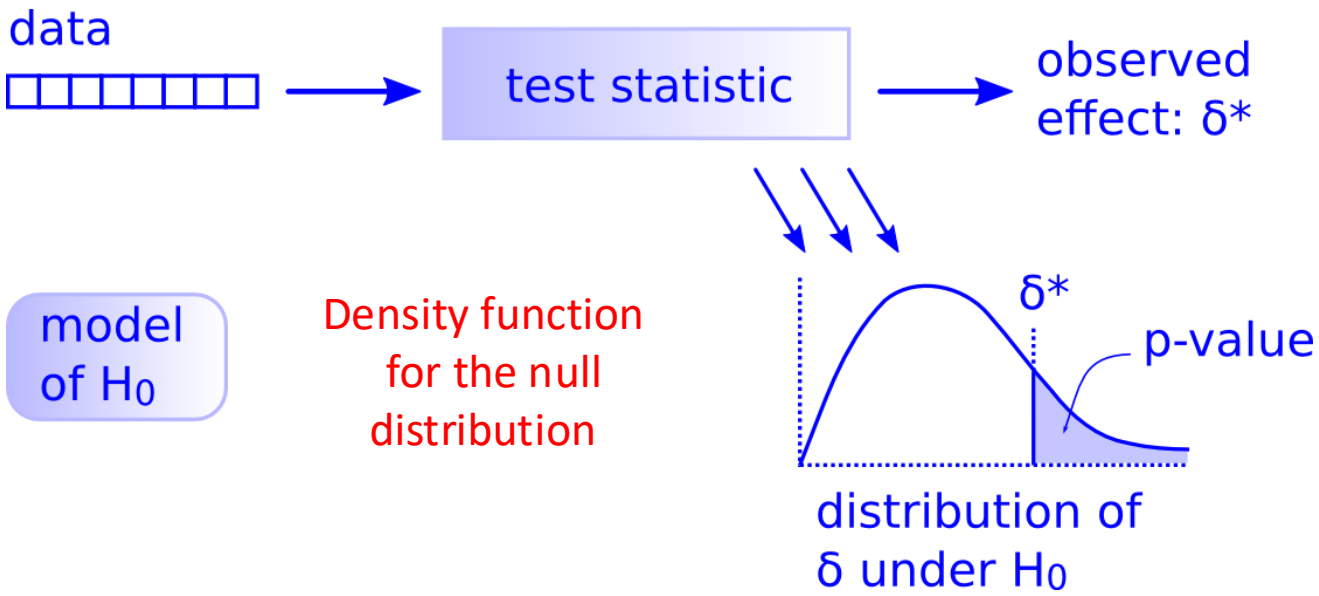
- Your account of the YCRC RStudio server will be deleted ~one month after the end of the semester
- You should be able to create a zip file of all your class material using:

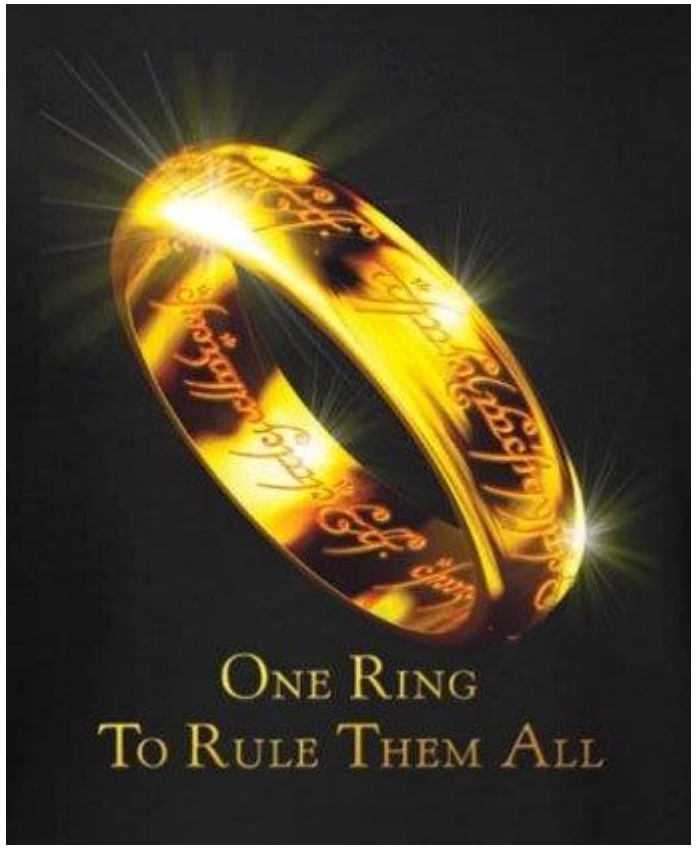
```
Sys.setenv("R_ZIPCMD"="zip")

SDS1000::save_zip_of_all_class_material()
```

Review and continuation of one-way analysis of variance (ANOVA)

One test to rule them all

There is only one hypothesis test!





Just follow the 5 hypothesis tests steps!

One-way ANOVA

An Analysis of Variance (ANOVA) is a parametric hypothesis test that can be used to examine if a set of means are all the same

H_0: $\mu_1 = \mu_2 = \ldots = \mu_k$

H_A: $\mu_i \neq \mu_j$ for some i, j

The statistic we use for a one-way ANOVA is the F-statistic

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

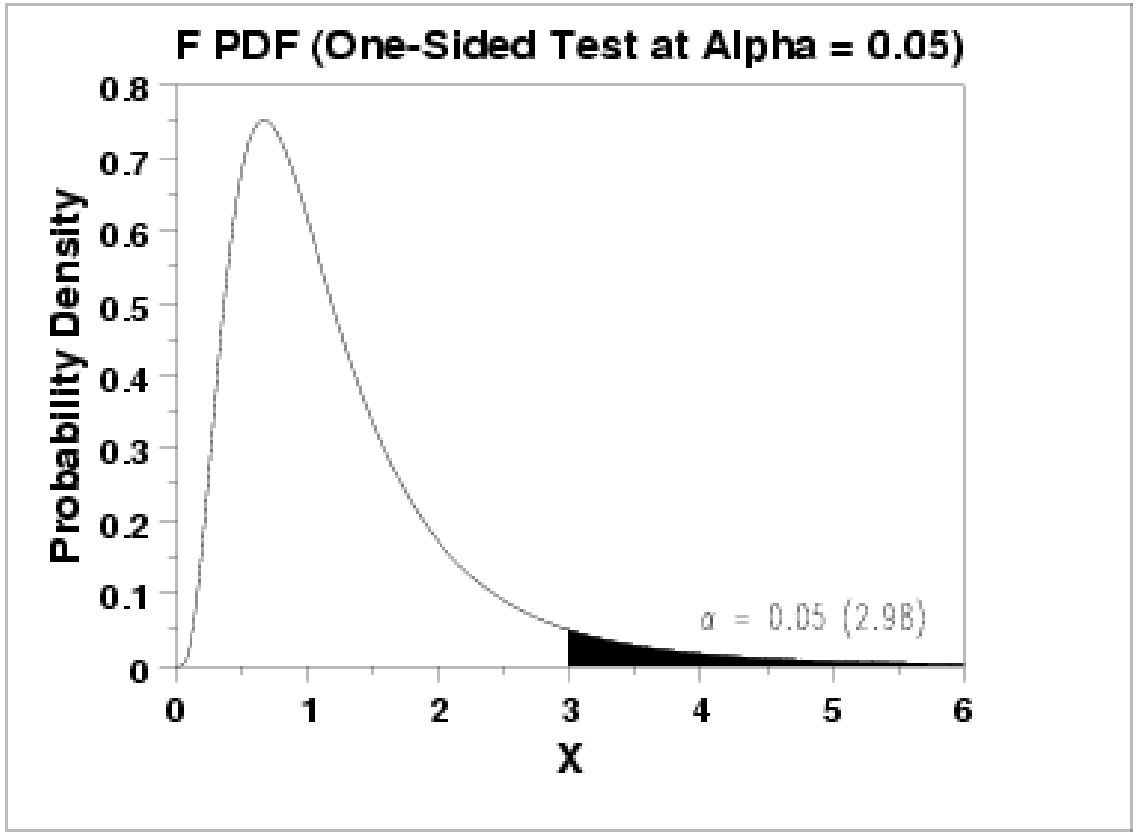
One-way ANOVA – the central idea

If H_0 is true, the F-statistic will come from an F distribution with parameters

- $df_1 = K - 1$
- $df_2 = N - K$

The F-distribution is valid if these conditions are met:

- The data in each group should follow a normal distribution
- The variances in each group should be approximately equal

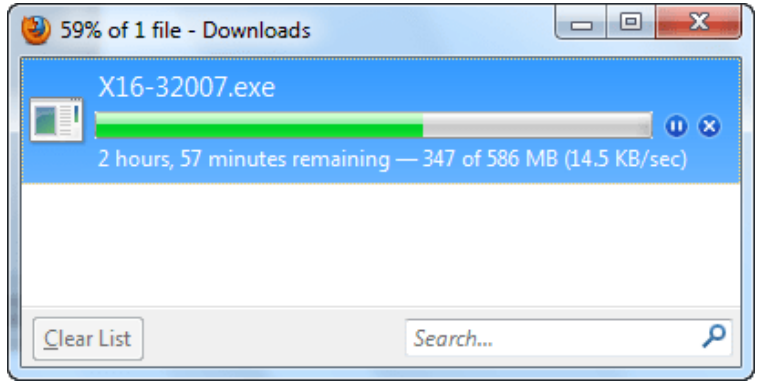


How does the time of the day affect download speeds?

To see whether the time of day affected the download speeds, a college sophomore performed an experiment

He placed a file on a remote server and then proceeded to download it at three different time periods of the day (7AM, 5PM, 12AM)

He downloaded the file 48 times in all, 16 times at each time of day, and recorded the time in seconds that the download took



1. State the null and alternative hypotheses

H_0: $\mu_{7AM} = \mu_{5PM} = \mu_{12AM}$

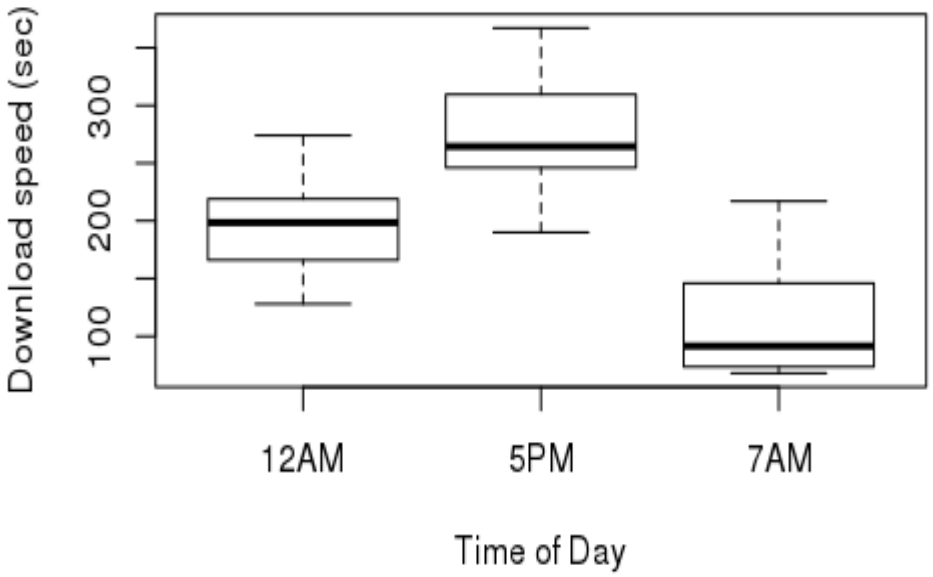
H_A: $\mu_i \neq \mu_j$ for one pair of the times of day

Let's check if the ANOVA conditions are met first…

Checking ANOVA conditions ('assumptions')

We can check if the data in each group is relatively normal by creating boxplots and seeing:

- Is the data very skewed?
- Are there are many outliers?



We can check the equal variance condition by seeing if the ratio of the largest to smallest standard deviation is greater than 2

- $s_{max}/s_{min} < 2$

$s_{7AM} = 47.6$ $s_{12AM} = 40.9$

$s_{5PM} = 52.2$

2. Calculating the observed F-statistic

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

K: the number of groups

N: total number of points

$\bar{x}_{tot}$: the mean across all the data

$\bar{x}_i$: the mean of group i

n_i: the number of points in group i

x_{ij} : the jth data point from group i

K = 3 different times of day

N = 48 total downloads (16 * 3)

$\bar{x}_{tot}$: the mean speed across all data

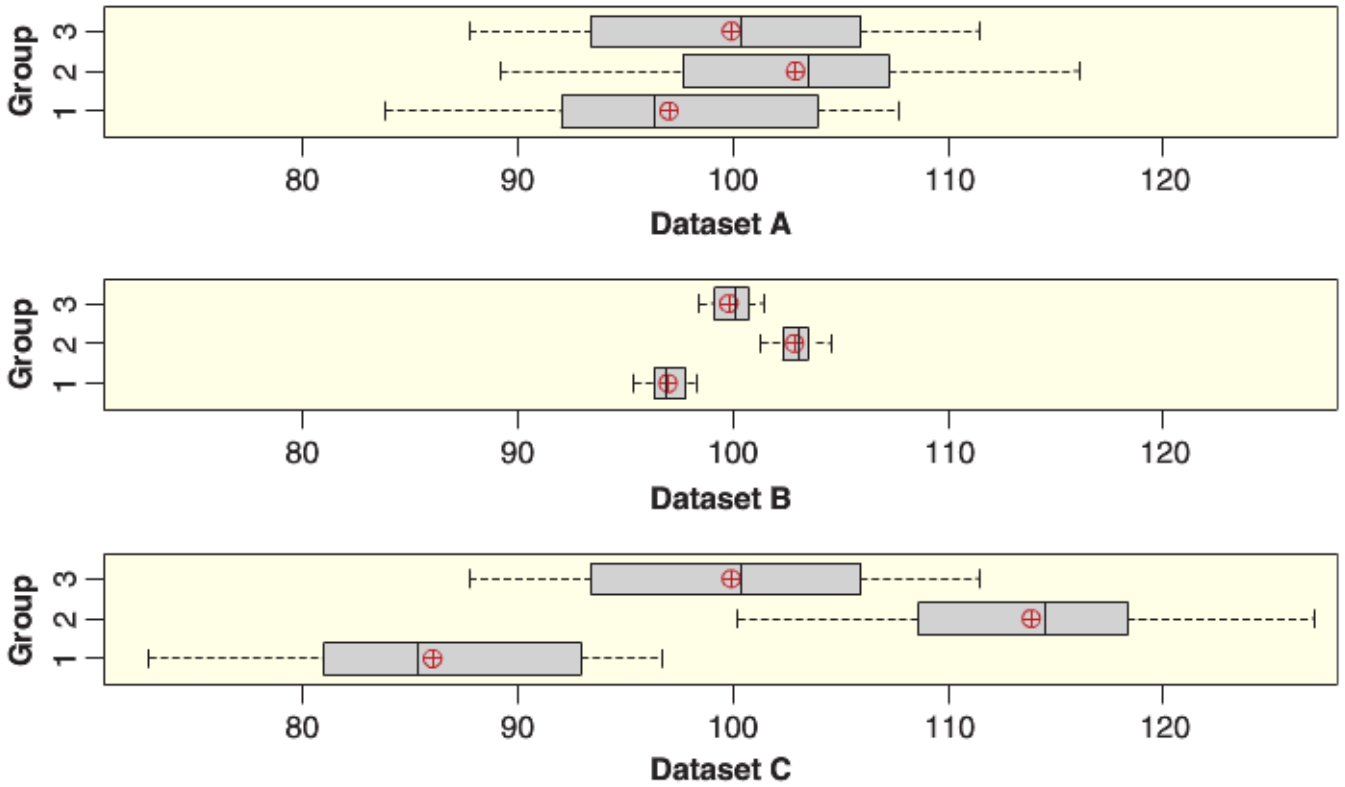
$\bar{x}_i$: the means for the ith time of day

n_i = 16 downloads for each time of day

x_{ij} : the jth download at the ith time of day

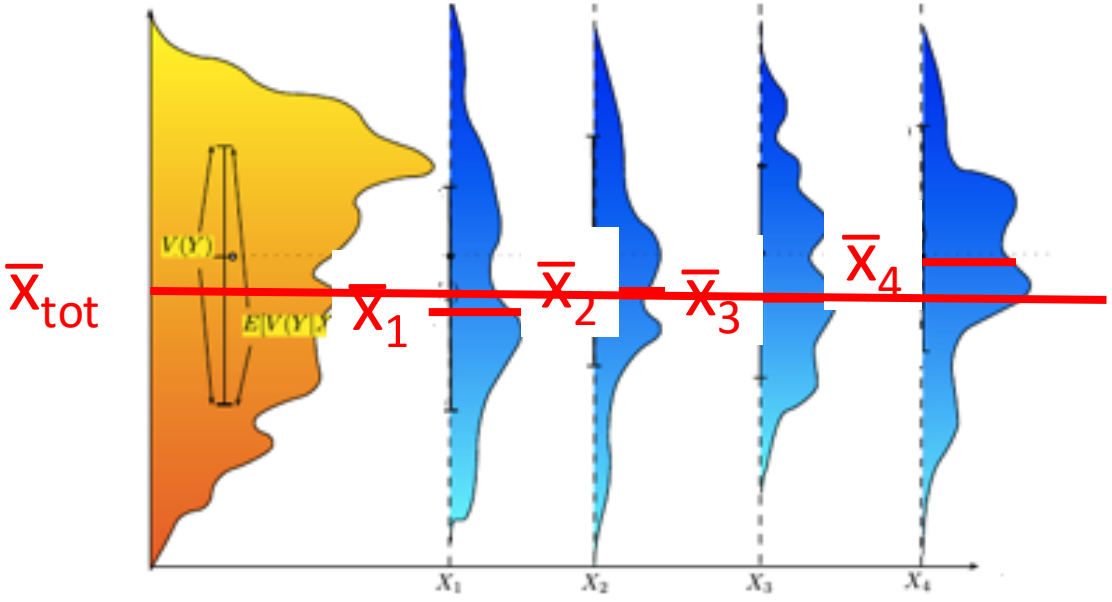
Why use the F-Statistic?

Which dataset gives the strongest evidence that there is a difference in population means?



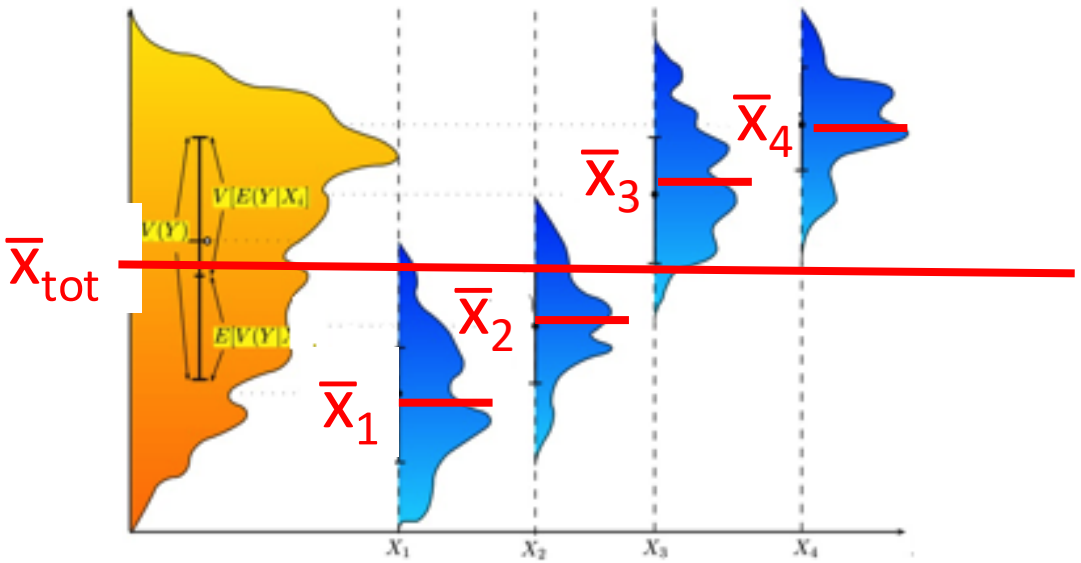
The F-Statistic

If H_0 is true, the data from all groups have the same means



- Similar means $\bar{x}_i$
- Similar spread s_i

If H_0 is not true, the data from all groups do not have the same mean



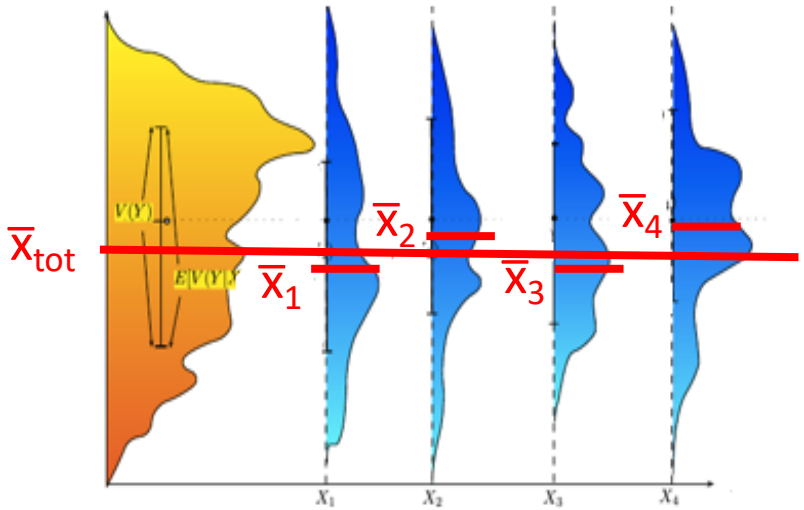
- Different means $\bar{x}_i$
- Smaller spreads s_i

The F-statistic

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

The F statistic measures a fraction of:

$$F = \frac{\text{variability between group means}}{\text{variability within each group}}$$



The F-statistic

Sum of Squares Group (SSG)

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

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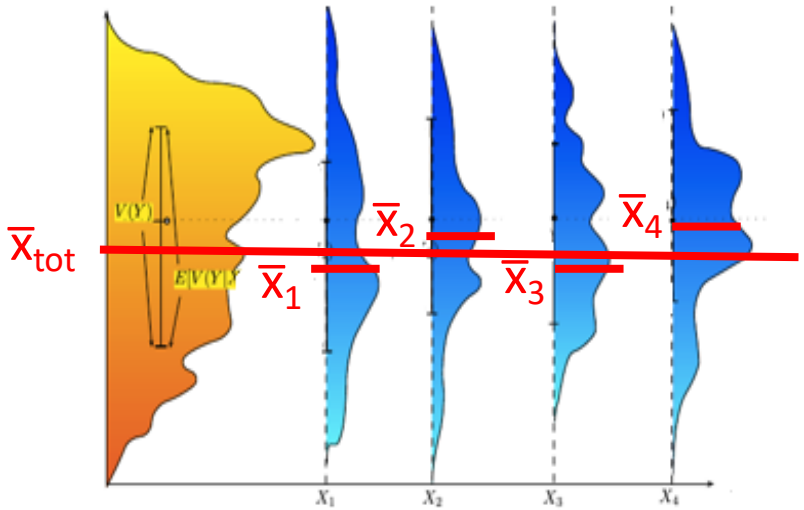

The F-statistic

Mean Squares Group (MSG)

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

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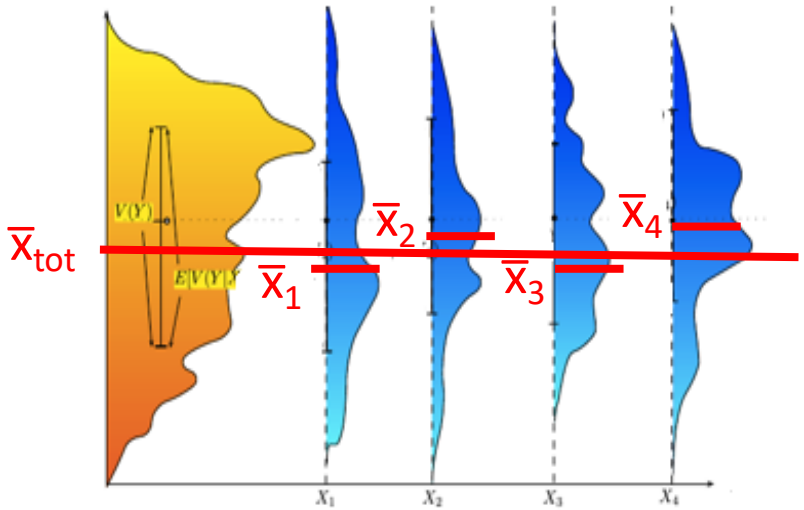


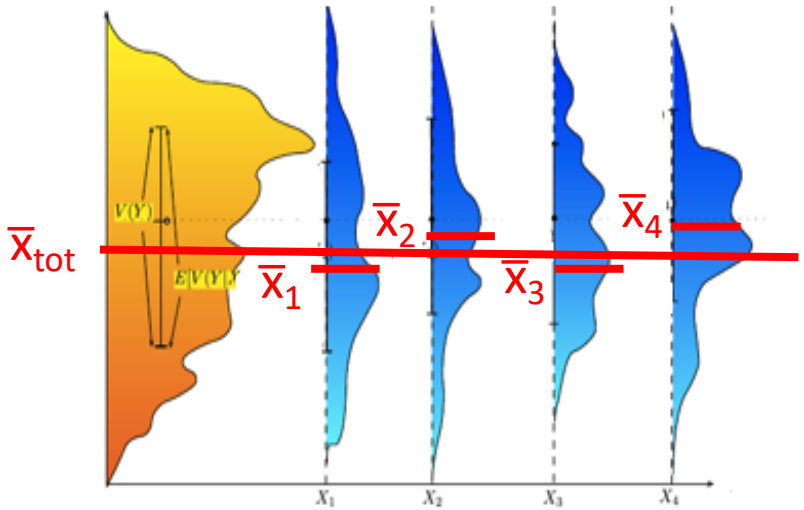
The F-statistic

Sum of Squares Error (SSE)

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

The F statistic measures a fraction of:

$$F = \frac{\text{Mean Squares Group (MSG)}}{\text{variability within each group}}$$



The F-statistic

Mean of Squares Error (MSE)

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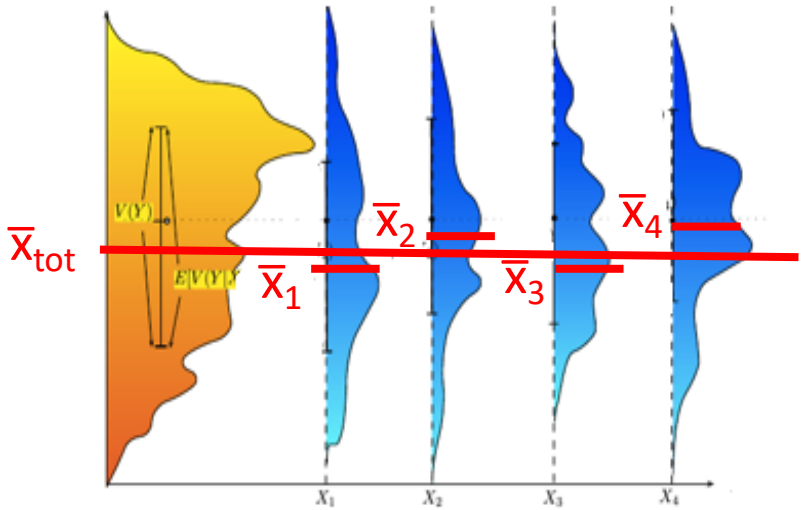

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The F statistic measures a fraction of:

$$F = \frac{\text{Mean Squares Group (MSG)}}{\text{Mean of Squares Error (MSE)}}$$

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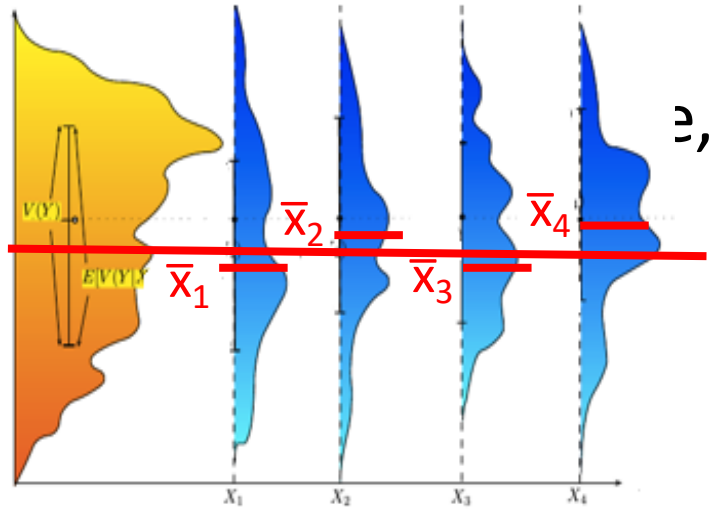


ANOVA table

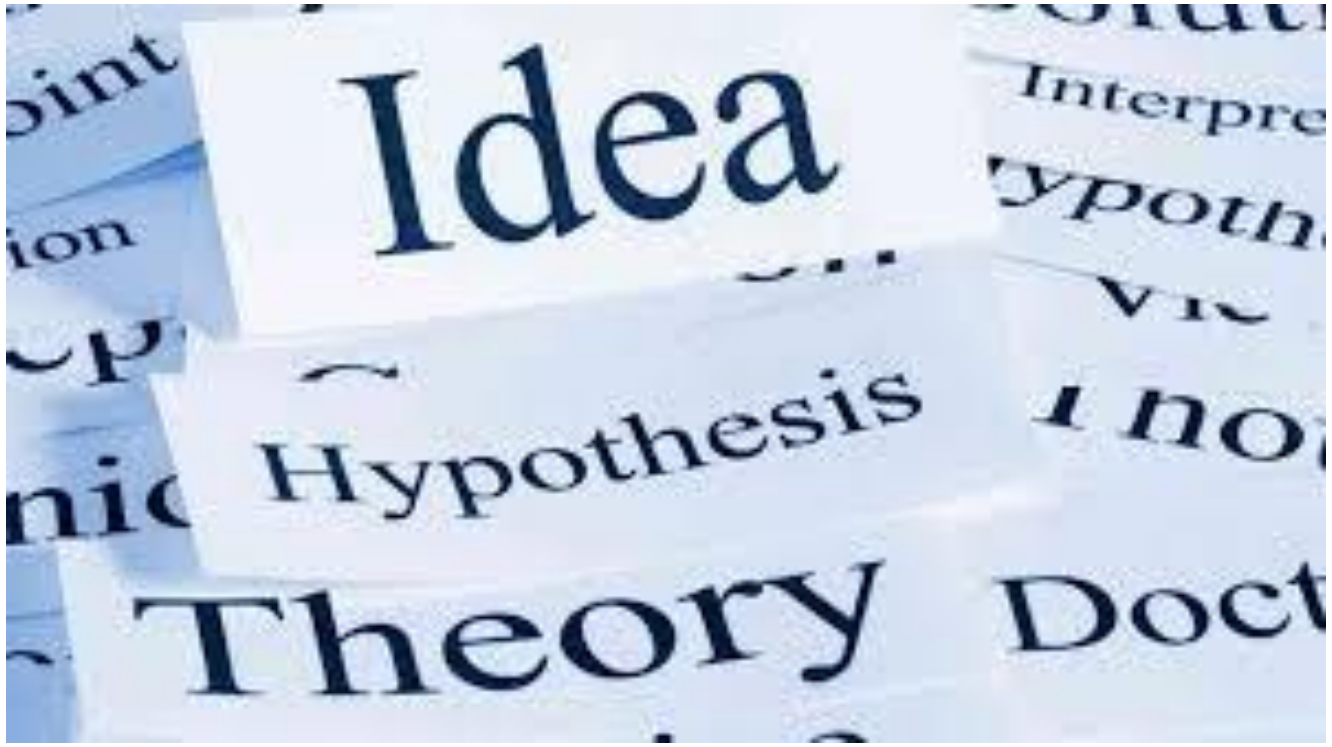
Source	df	Sum of Sq.	Mean Square	F-statistic	p-value
Groups	$k-1$	SSG	$MSG = \frac{SSG}{k-1}$	$F = \frac{MSG}{MSE}$	Upper tail $F_{k-1,n-k}$
Error	$n-k$	SSE	$MSE = \frac{SSE}{n-k}$		
Total	$n-1$	$SSTotal$			

$$F = \frac{\text{between-group variability}}{\text{within-group variability}} = \frac{\frac{1}{K-1}\sum_{i=1}^{K} n_i(\bar{x}_i - \bar{x}_{tot})^2}{\frac{1}{N-K}\sum_{i=1}^{K}\sum_{j=1}^{n_i}(x_{ij} - \bar{x}_i)^2}$$

Try analyzing the download speed data at home!

Theories of null significance hypothesis testing

Theories of hypothesis tests



Two theories of hypothesis testing

Null-hypothesis significance testing (NHST) is a hybrid of two theories:

1. Significance testing of Ronald Fisher
2. Hypothesis testing of Jezy Neyman and Egon Pearson

Fisher (1890-1962)

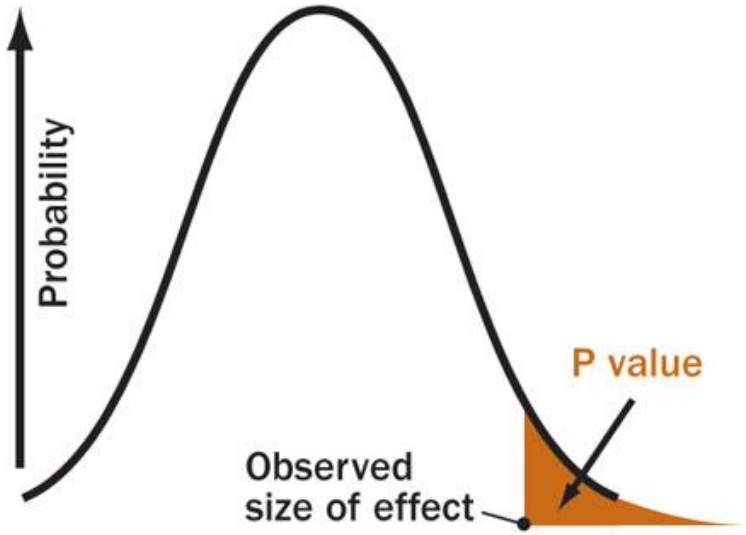
Neyman (1894-1981)

Pearson (1895-1980)

Ronald Fisher's significance testing

Views the p-value as strength of evidence against the null hypothesis

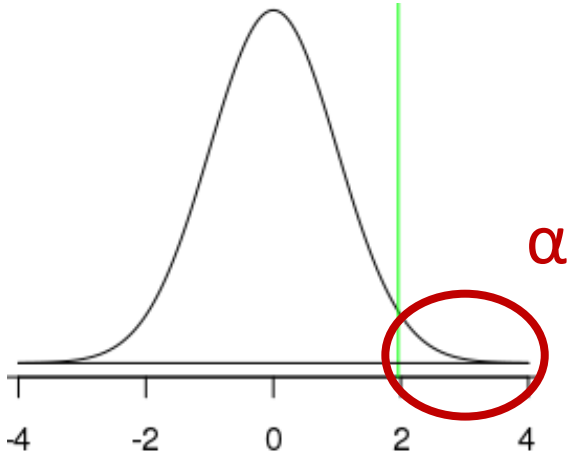
- P-values part of an on-going scientific process: tells the experimenter "what results to ignore"



Neyman-Pearson null hypothesis testing

Makes ***a formal decision*** in statistical tests

Null distribution



Reject H_0: if the observed sample statistic is beyond a fixed value

- i.e., reject H_0 if the p-value is less than some predetermined **significance level** α

Do not reject H_0: if the observed sample statistic is not beyond a fixed value. This means the test is inconclusive

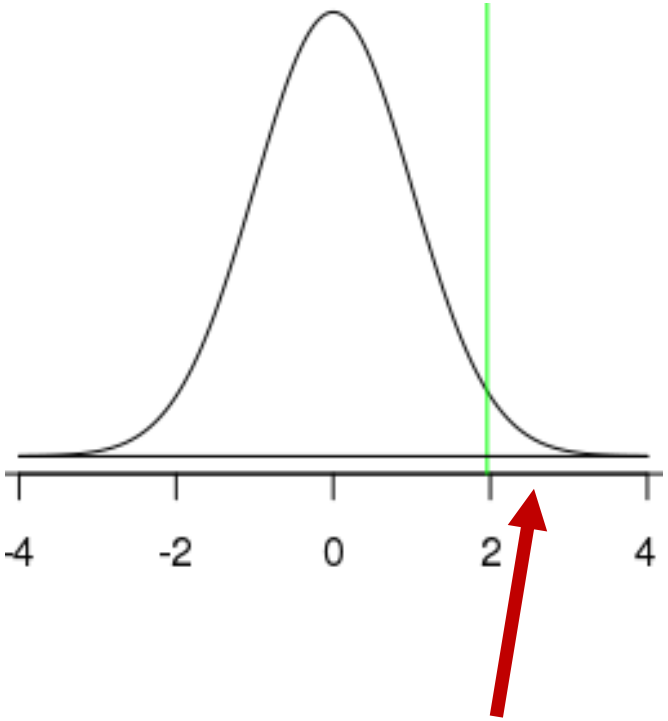
Neyman-Pearson frequentist logic

Type I error: incorrectly rejecting the null hypothesis when it is true

When the null hypothesis is true, only ~5% of the time will we falsely reject it (for α = 0.05)

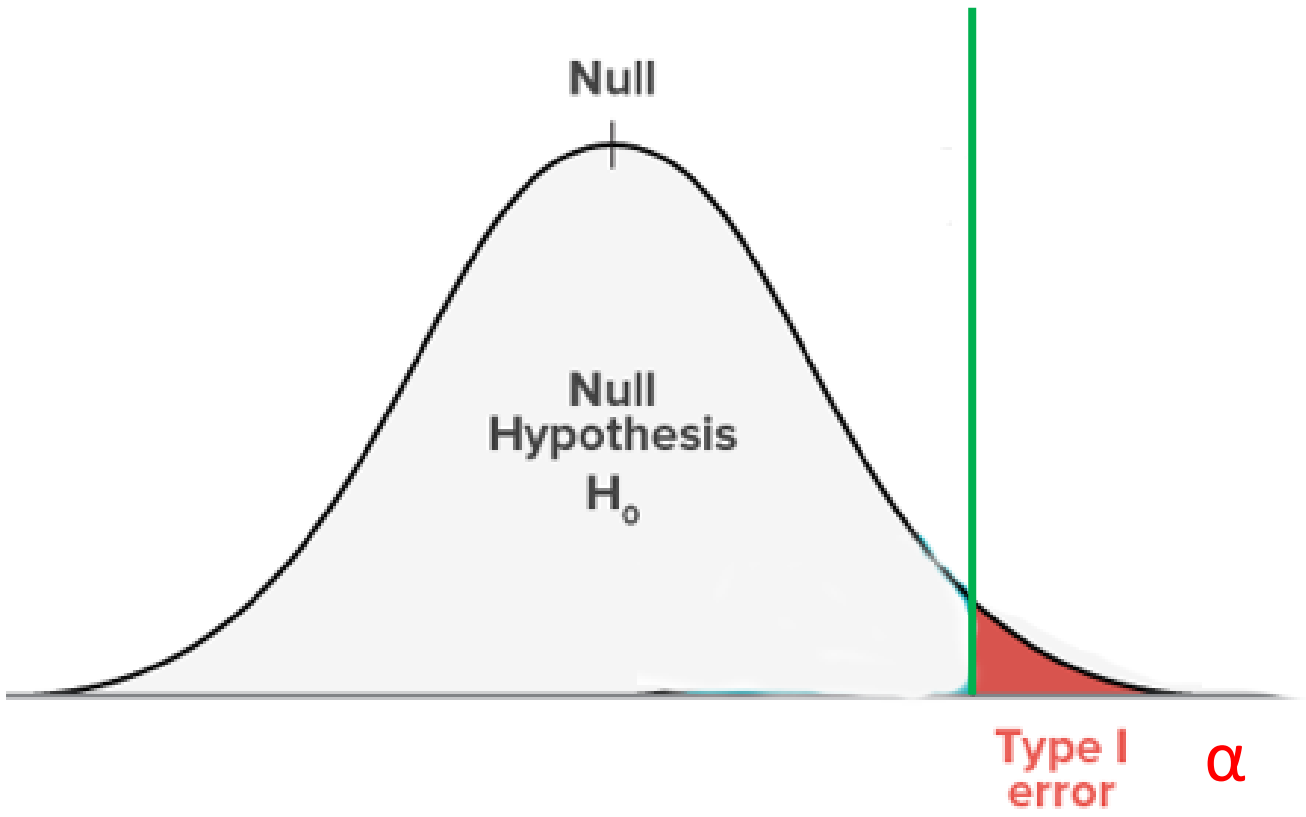
- i.e., we only make type I errors 5% of the time

Null distribution

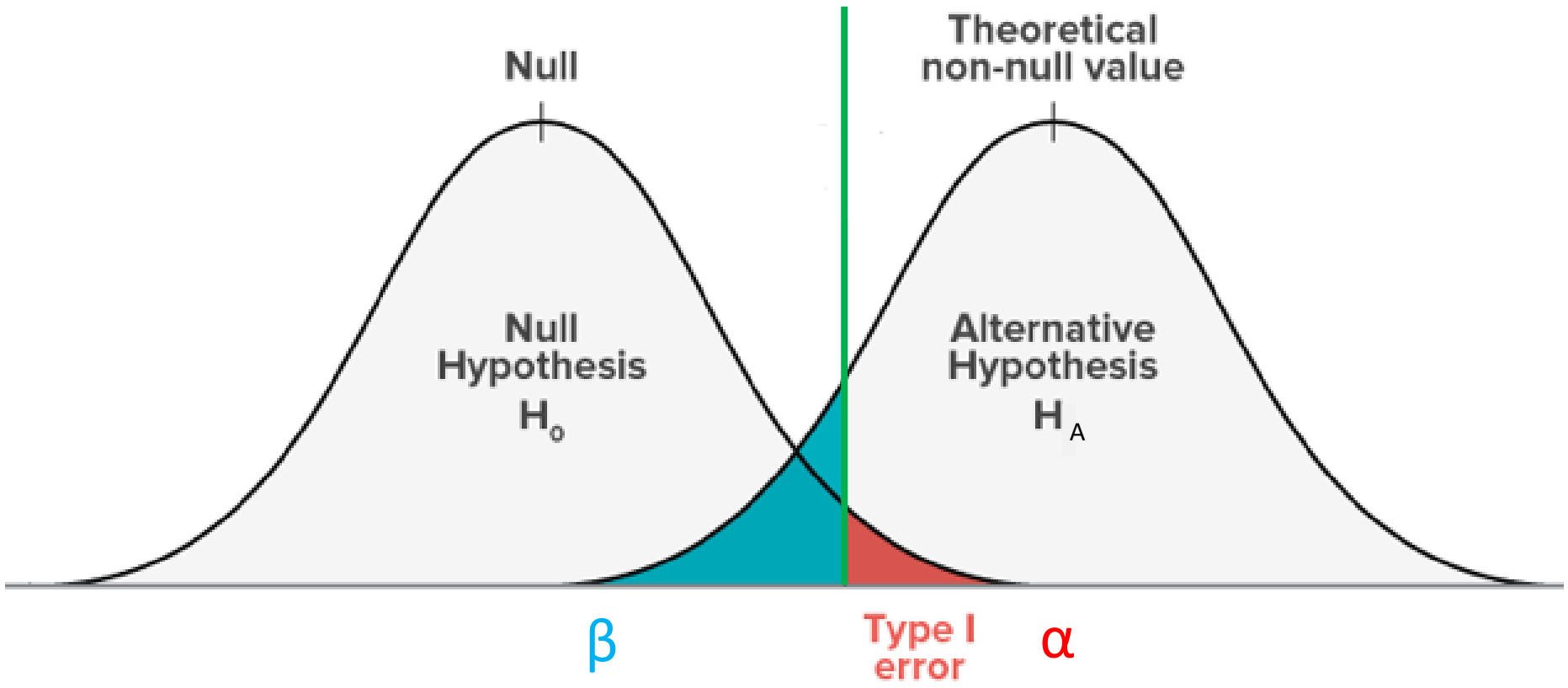


The null distribution is true but statistic landed here

Neyman-Pearson Frequentist logic



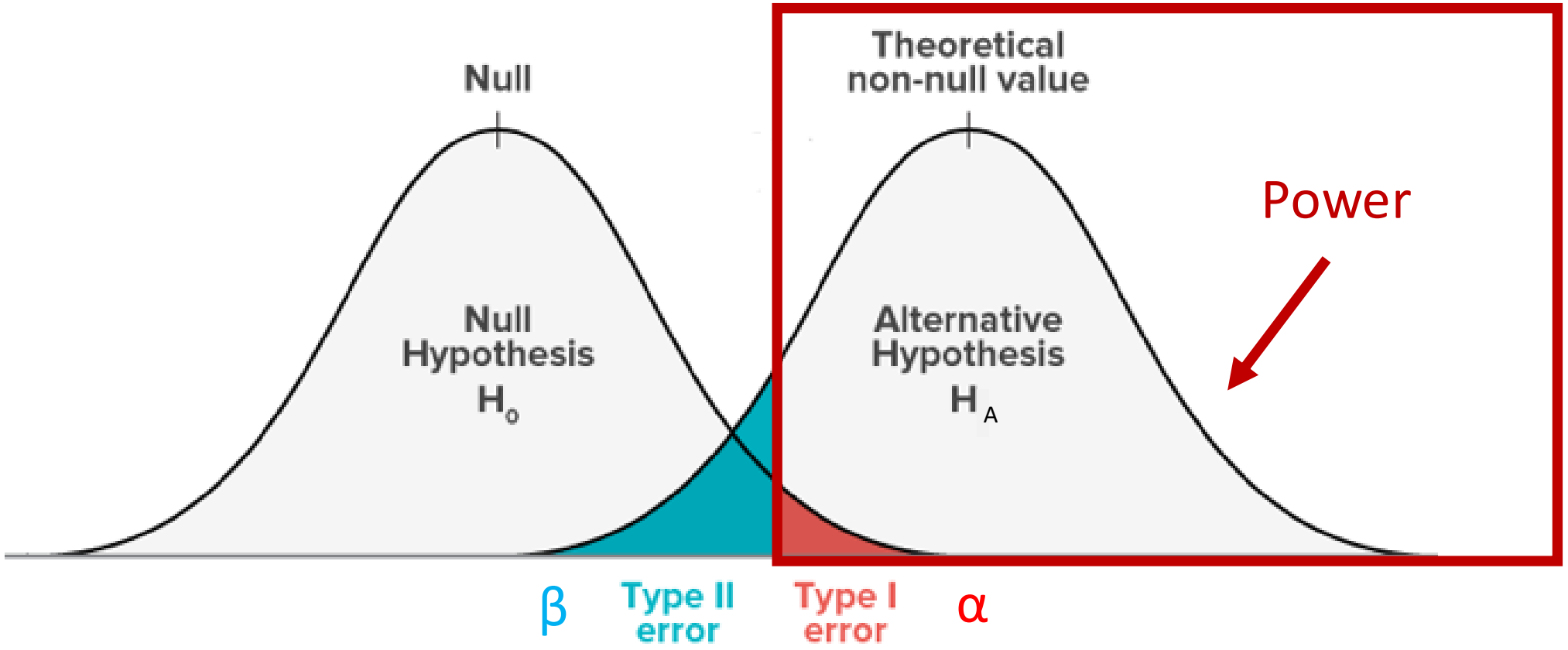
Neyman-Pearson Frequentist logic



Type II error: incorrectly failing to reject H_0 when it is false

- The rate at which we make type II errors is often denoted with the symbol β

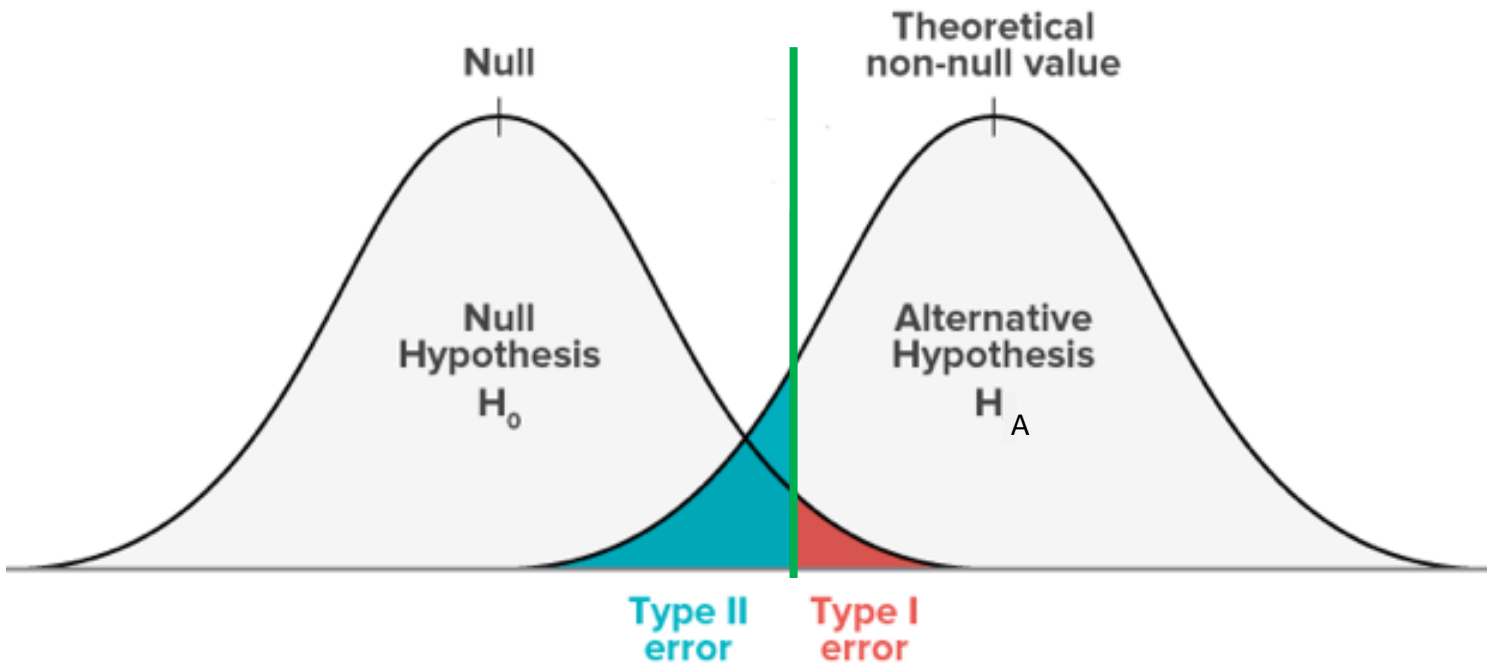
Neyman-Pearson Frequentist logic



The **power** of a test is the probability we reject the H_0 when it is **false**

- 1 - β
- For a fixed α level, it would be best to use the most powerful test

Type I and Type II Errors



	H_0 is true	H_A is true (H_0 is false)
Reject H_0	Type I error (α) (false positive)	No error
Do not reject H_0	No error	Type II error (β) (false negative)

Problems with the NP hypothesis tests

<u>Problem 1</u>: we are interested in the results of a specific experiment, not whether we are right most of the time

- E.g., 95% of these statements are true:
  - Calcium is good for your heart, Paul is psychic, Buzz and Doris can communicate, …

<u>Problem 2</u>: Arbitrary thresholds for alpha levels

- P-value = 0.051, we don't reject H_0

Problems with the NP hypothesis tests

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<u>Problem 2</u>: Arbitrary thresholds for alpha levels

- P-value = 0.051, we don't reject H_0?

<u>Problem 3</u>: running many tests can give rise to a high number of type I errors

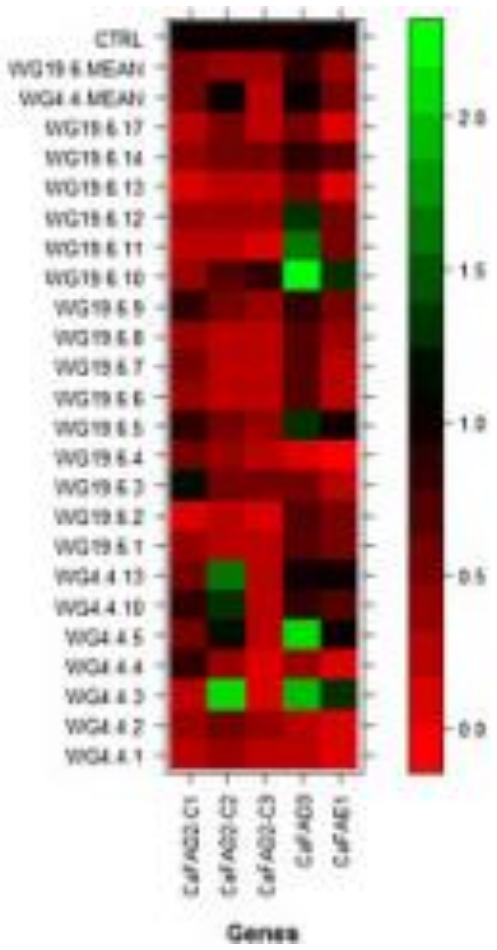
Genes and leukemia example

Scientists collected 7129 gene expression levels from 38 patients to find genetic differences between two types leukemia (L1 and L2)

Suppose there was no difference in any of the gene expression levels for the two types of leukemia

- H_0: $\mu_{L1} = \mu_{L2}$ is true for all genes

Q: If each gene was tested separately using a significance level of $\alpha = 0.05$, approximately how many type I errors would be expected?



Side note: correcting for multiple comparisons

There are methods that try to correct for running multiple hypothesis tests

The ***Bonferroni correction*** is one simple way to control the probability of ***any*** hypothesis test giving a type I error

- i.e., controls the familywise error rate (no type I errors for any of the tests run)

It works by dividing the initial α level by the number of tests run

- E.g., $\alpha = 0.05/7129 = 0.000007$
- All p-values need to be below this level to be considered statistically significant
- This can lead to many type II errors (Type II error: failure to reject H_0 when it is false)

The problem of multiple testing

For $\alpha = 0.05$, ~5% of all published research findings should incorrectly reject the null hypothesis

Publication bias (file drawer effect): Generally positive results are more likely to be published, so if you read the literature, the proportion of incorrect results could be greater than 5%

Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

The Earth Is Round ($p < .05$)

Jacob Cohen

After 4 decades of severe criticism, the ritual of null hypothesis significance testing—mechanical dichotomous decisions around a sacred .05 criterion—still persists. This article reviews the problems with this practice, including

sure how to test H_0, chi-square with Yates's (1951) correction or the Fisher exact test, and wonders whether he has enough power. Would you believe it? And would you believe that if he tried to publish this result without a

American Statistical Association's Statement on p-values

Some thoughts…

Better to have hypothesis tests than none at all. Just need to think carefully and use your judgment.

Report effect size in most cases – i.e., confidence intervals

Report the p-values rather than accept/reject H_0

- i.e., report $p = 0.023$ not $p < 0.05$

Replicate findings (perhaps in different contexts) to make sure you get the same results

Be a good/honest scientists and try to get at the Truth!

Parametric inference for regression

Review of regression (class 6 and 7)

In **linear regression** we fit a <u>line</u> to the data, called the **regression line**

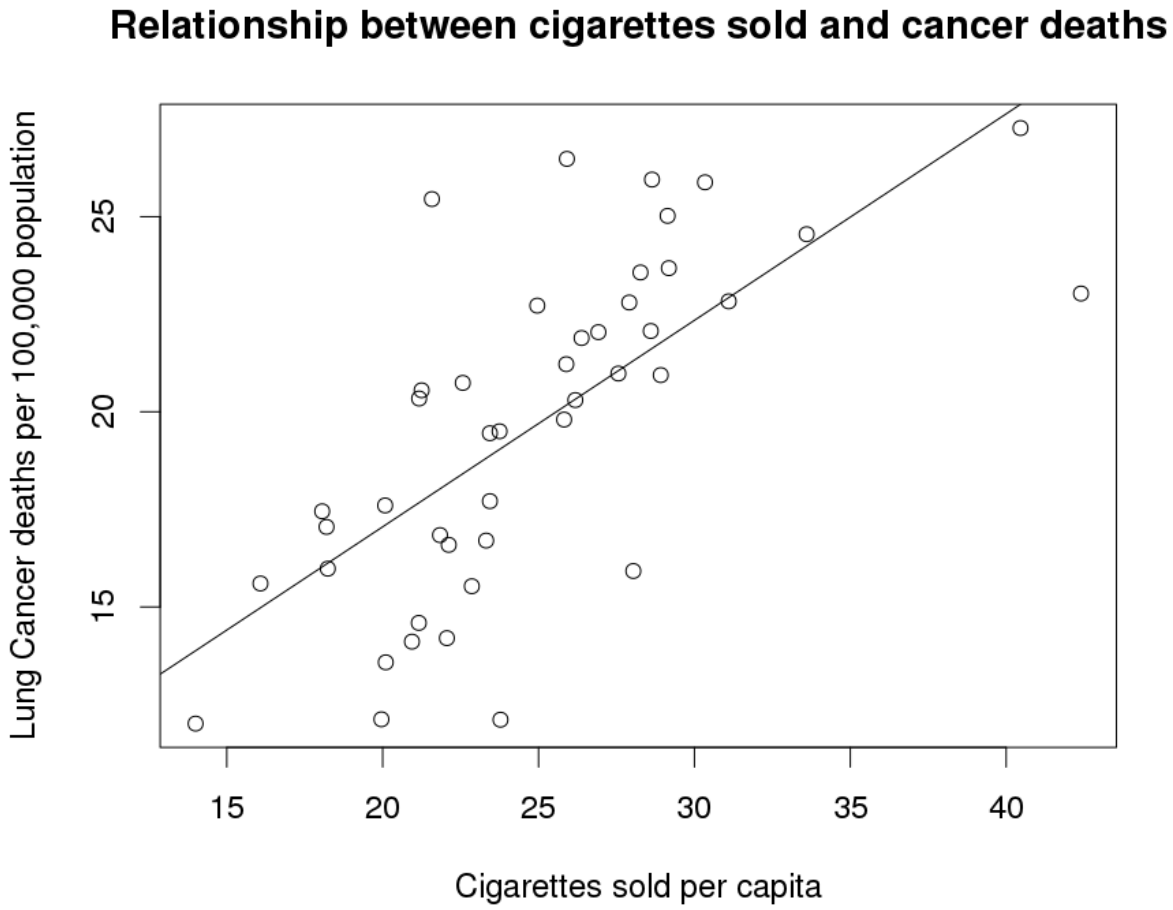
$$\hat{y} = a + b \cdot x$$

$$\textit{Predicted response} = a + b \cdot \textit{Explanatory}$$

Change in notation to be consistent with the Lock5 and what most statisticians use

$$\textit{Predicted response} = b_0 + b_1 \cdot \textit{Explanatory}$$

Review cancer smoking regression line



$$\hat{y} = b_0 + b_1 \cdot x$$

```
R: my_fit <- lm(y ~ x)
   coef(my_fit)
```

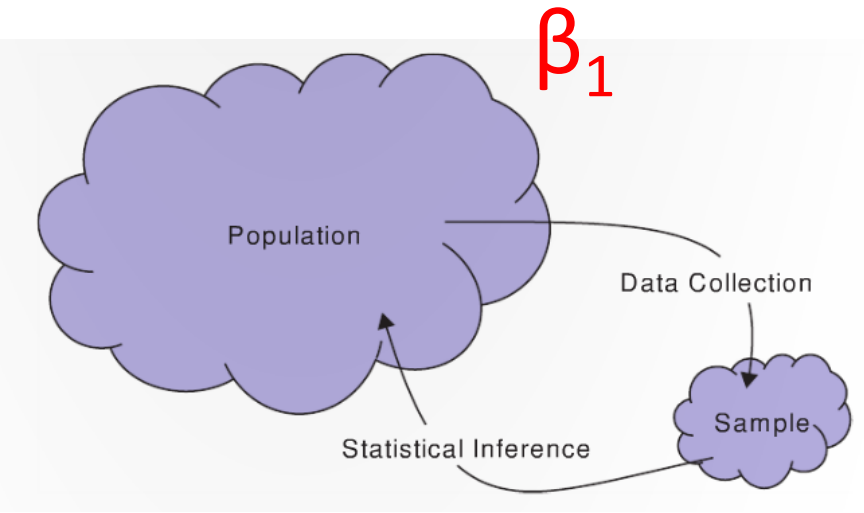
$b_0 = 6.47 \quad b_1 = 0.53$

$$\hat{y} = 6.47 + .53 \cdot x$$

Inference on simple linear regression

The Greek letter β_1 is used to denote the slope of the population

The letter b_1 is typically used to denote the slope of the sample

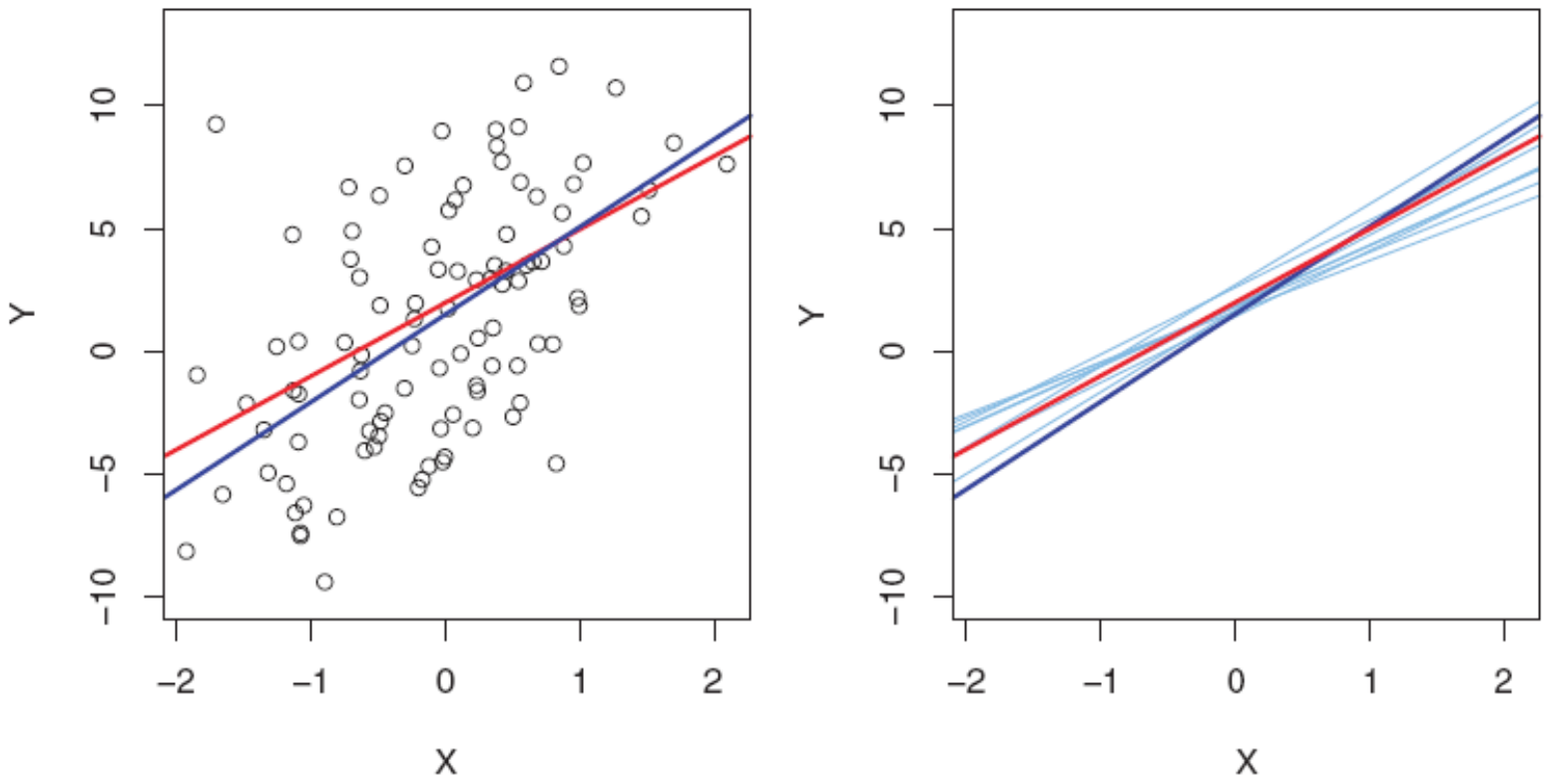


β_1

b_1

Population: β_1

Sample estimates: b_1



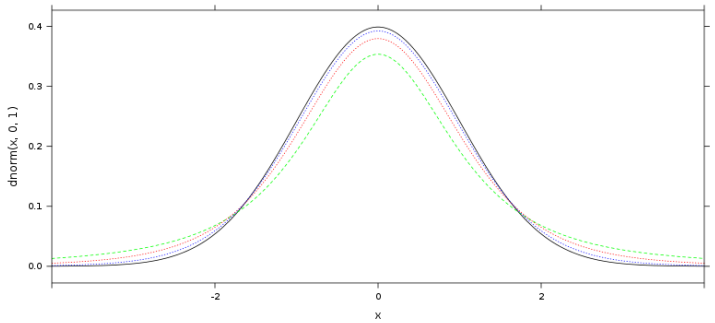
Hypothesis test for regression coefficients

We can run hypothesis tests to assess whether there is a linear relationship between y and x

- H_0: $\beta_1 = 0$ (slope is 0: no linear relationship between x and y
- H_A: $\beta_1 \neq 0$

One type of hypothesis test we can run is based on a t-statistic: $t = \dfrac{b_1 - 0}{SE_{b_1}}$

- The t-statistic comes from a t-distribution with n - 2 degrees of freedom

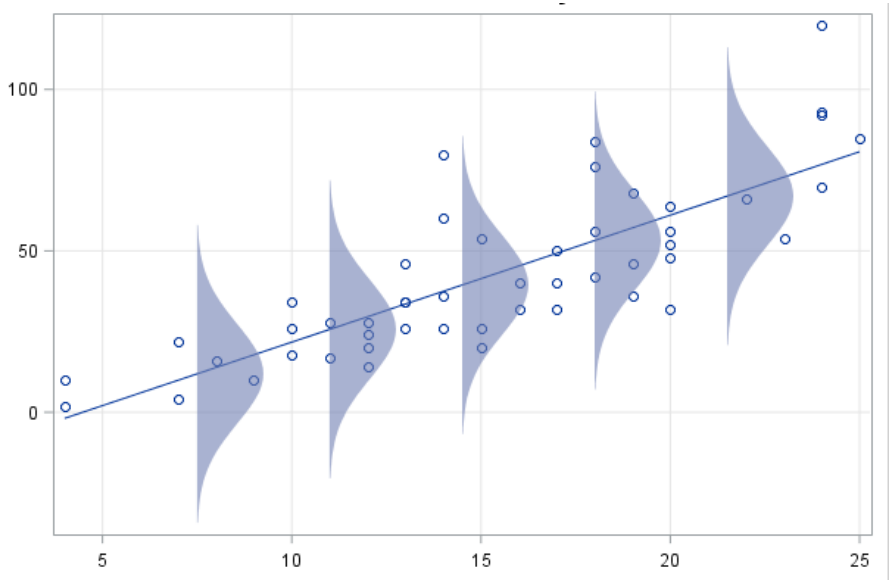


$$SE_{b_1} = \sqrt{\frac{1}{n-2}\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$$

Inference using parametric methods

When using parametric methods, we make the following (LINE) assumptions:

- **Linearity**: A line can describe the relationship between x and y
- **Independence**: each data point is independent from the other points
- **Normality**: errors are normally distributed
- **Equal variance (homoscedasticity)**: constant variance of errors over the whole range of x values



These assumptions are usually checked after the models are fit using 'regression diagnostic' plots.

Confidence intervals for regression coefficients

For the slope coefficient , the confidence interval is: $b_1 \pm t^* \cdot SE_{b_1}$

Where: $SE_{b_1} = \sqrt{\frac{1}{n-2}\frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}}$

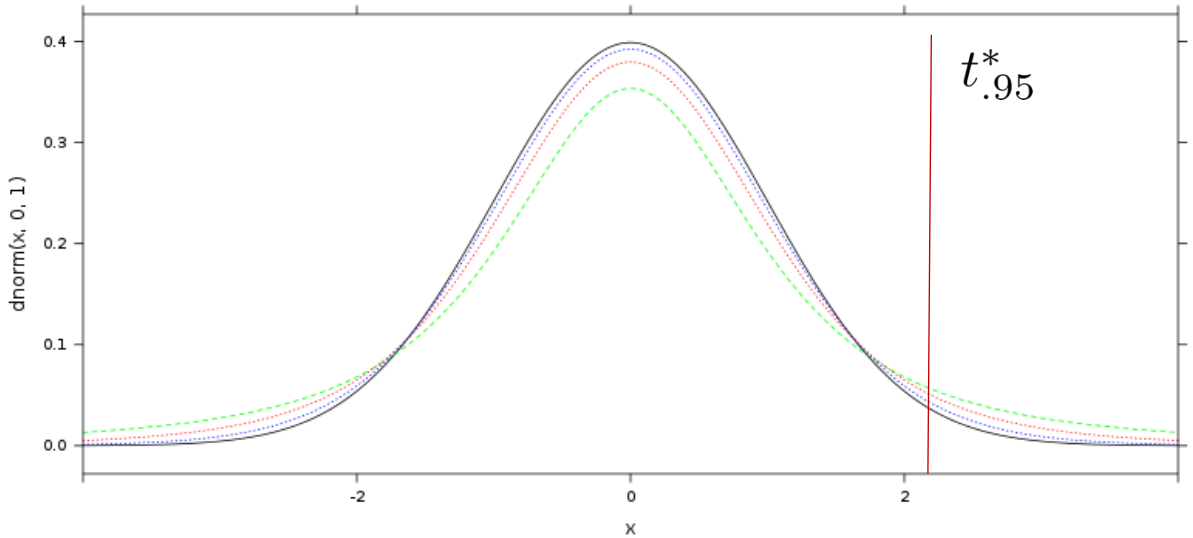
t* is the critical value for the t_{n-2} density curve needed to obtain a desired confidence level

N(0, 1)

df = 2

df = 5

df = 15



Let's try it in R!

Bonus material

Inference in regression using simulation methods

Review of regression (class 6 and 7)

In **linear regression** we fit a <u>line</u> to the data, called the **regression line**

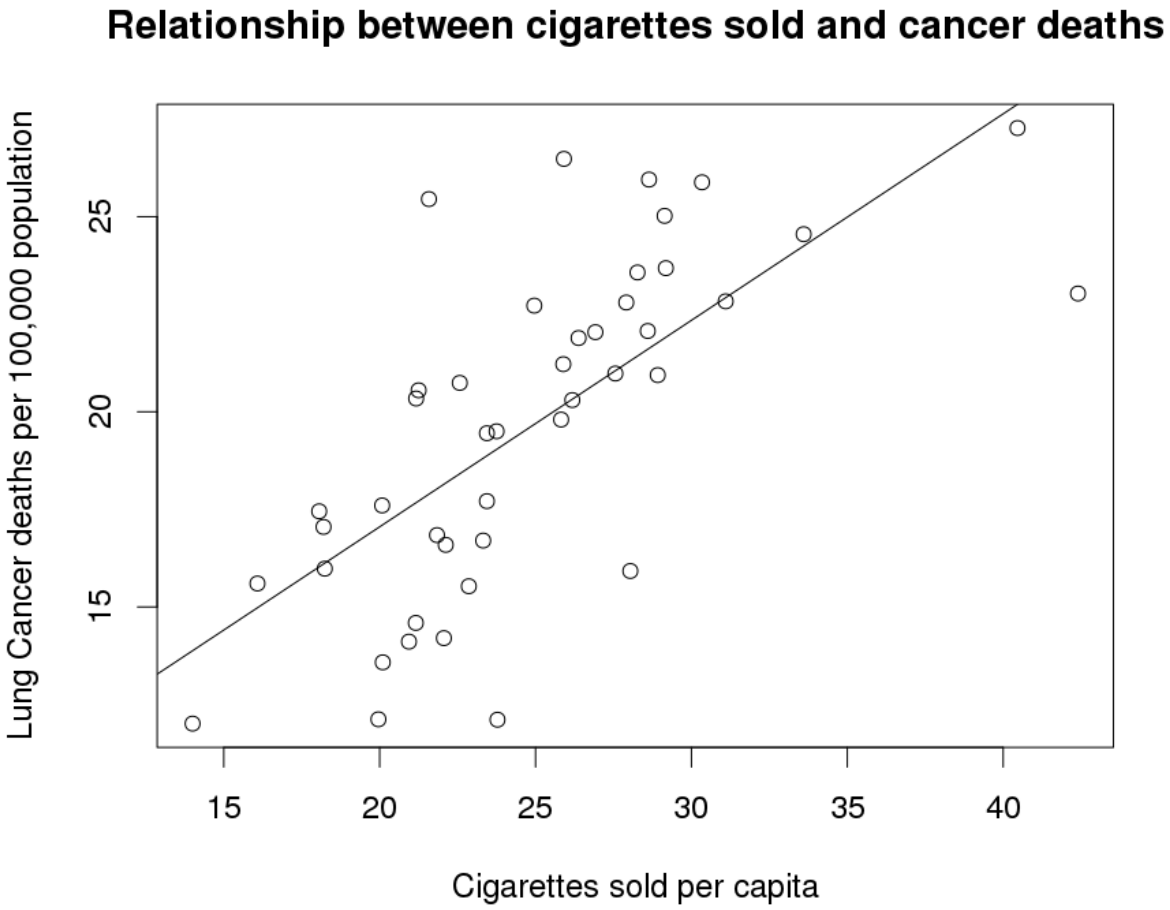
$$\hat{y} = a + b \cdot x$$

$$\textit{Predicted response} = a + b \cdot \textit{Explanatory}$$

Change in notation to be consistent with the Lock5 and what most statisticians use

$$\textit{Predicted response} = b_0 + b_1 \cdot \textit{Explanatory}$$

Review cancer smoking regression line



$$\hat{y} = b_0 + b_1 \cdot x$$

```
R: my_fit <- lm(y ~ x)
   coef(my_fit)
```

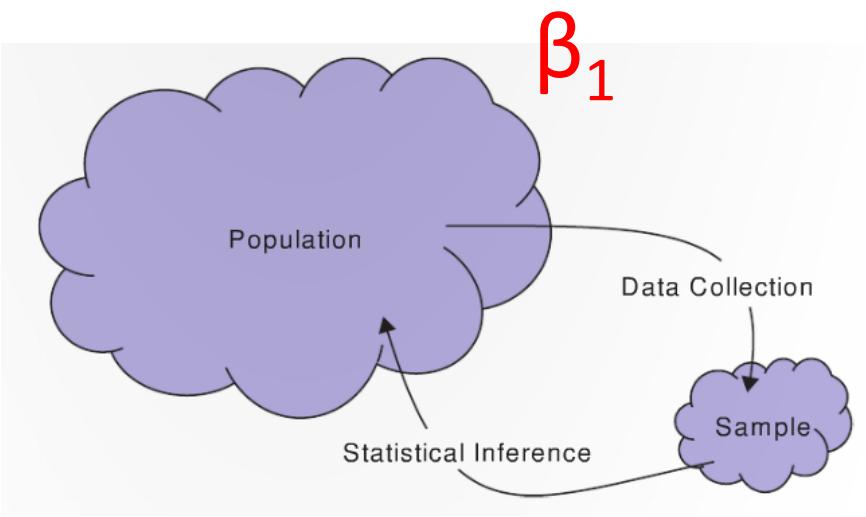
$b_0 = 6.47 \quad b_1 = 0.53$

$$\hat{y} = 6.47 + .53 \cdot x$$

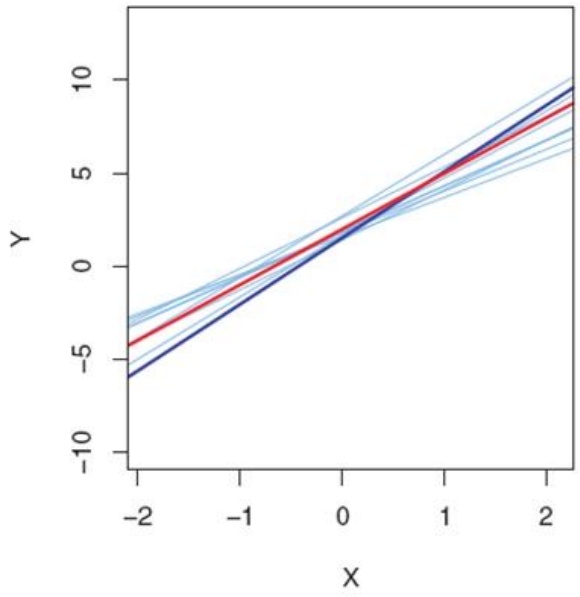
Inference on simple linear regression

The Greek letter β_1 is used to denote the slope of the population

The letter b_1 is typically used to denote the slope of the sample



b_1



Using the bootstrap to create confidence intervals

We can use the bootstrap to create confidence intervals by:

1. Creating a bootstrap sample by sampling with replacement from our *paired data*
   - SDS1000: resample_pairs(v1, v2)
2. Fitting a regression line to our bootstrap sample and extracting the slope b
3. Repeat 10,000 times to get a bootstrap distribution of b's
4. Taking the standard deviation of the bootstrap distribution to get SE*
5. Using our confidence interval formula:

$$\textit{Statistic} \pm 1.96 \cdot SE^*$$

State	Cig per capita	Lung
AL	18.2	17.05
AZ	25.82	19.8
AR	18.24	15.98
CA	28.6	22.07
CT	31.1	22.83
DE	33.6	24.55
DC	40.46	27.27

Using permutation hypothesis tests

If we wanted to run a hypothesis tests for the regression slope, how would we write the null and alternative hypotheses using symbols?

H_0: $\beta = 0$

H_A: $\beta \neq 0$

Any ideas how to run a permutation test to assess whether H_0: $\beta = 0$?

State	Cig per capita	Lung
AL	18.2	17.05
AZ	25.82	19.8
AR	18.24	15.98
CA	28.6	22.07
CT	31.1	22.83
DE	33.6	24.55
DC	40.46	27.27

Using permutation hypothesis tests

We could use run a permutation test for H_0: $\beta = 0$ by creating a null distribution using:

1. Shuffle one of the columns of data
2. Fitting a regression line to our bootstrap sample and extracting the slope b
3. Repeat 10,000 times to get a null distribution of b's

We can obtain a p-value by seeing how many points in the null distribution are greater than the observed statistic value of b

Let's try it in R!

State	Cig per capita	Lung
AL	18.2	17.05
AZ	25.82	19.8
AR	18.24	15.98
CA	28.6	22.07
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